

Lecture 22: UCB-VI Algorithm

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In this lecture, we introduce the *upper confidence bound value iteration (UCB-VI)* algorithm for reinforcement learning [1]. This algorithm generalizes UCB from the multi-armed bandit to the reinforcement learning setting, allowing us to make efficient use of our exploration data. For simplicity, we will only consider the fixed horizon case with time-varying transitions and rewards and a fixed start state.

1 Notation

We will use the following terms to define and evaluate our algorithm.

$S = \mathcal{S} $	size of the state space
$A = \mathcal{A} $	size of the action space
H	horizon
K	number of episodes
$T = KH$	total number of iterations
δ	confidence parameter
$L = \log\left(\frac{SAT}{\delta}\right)$	component of exploration bonus
$\mathcal{D}$	dataset
$N : H \times \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{N}$	visitation count
$Q : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$	action-value function
$V : \mathcal{S} \rightarrow \mathbb{R}$	state-value function
$b : N \rightarrow \mathbb{R} = c\sqrt{\frac{H^2 L}{N}}$	exploration bonus
$\pi^k = \{\pi_1, \pi_2, \dots, \pi_H\}$	policy sequence

Remarks:

- We will often write random variables X as X_h^k . This represents the variable X at the timestep $h \in [H]$ (where H is the horizon), and the *expectation* over the K episodes.
- Note that π^k refers to a sequence of policies (one for each timestep), rather than a single policy. This is necessary because transition and reward functions can vary over time.

2 UCB-VI Algorithm

The upper confidence bound value iteration algorithm consists of two parts: **Value Iteration** and **Policy Execution**. Both processes are executed K times, once for each episode. Policy execution executes a policy that is greedy on the current Q values and adds newly encountered $(h, s_h, a_h, s_{h+1}, r_h(s_h, a_h))$ transitions to the dataset. Value iteration uses this dataset of empirical transitions to estimate the Q values of state-action pairs.

Algorithm 1: Upper Confidence Bound Value Iteration

Parameters: episodes K , confidence δ
Initialize: $\mathcal{D} = \emptyset$; $Q_h(s, a) = H \forall_{h \in [H], s \in \mathcal{S}, a \in \mathcal{A}}$; $V_{H+1}(s) = 0 \forall_{s \in \mathcal{S}}$
for episode $k \in [K]$ **do**
 1. Value Iteration
 for timestep $h = H, H - 1, \dots, 1$ **do**
 for $(s, a) \in \mathcal{S} \times \mathcal{A}$ **do**
 if $(s, a) \in \mathcal{D}$ **then**
 compute probability $\hat{P}_h(s'|s, a) = \frac{N_h(s, a, s')}{N_h(s, a)} \forall_{s' \in \mathcal{S}}$
 assign $Q_h(s, a) \triangleq \min\{H, r_h(s, a) + (\hat{P}_h V_{h+1})(s, a) + b(N_h(s, a))\}$
 end
 end
 for $s \in \mathcal{S}$ **do**
 set $V_h(s) = \max_{a \in \mathcal{A}} Q_h(s, a)$
 end
 2. Policy Execution
 for timestep $h \in [H]$ **do**
 take greedy action $a_h = \operatorname{argmax}_{a \in \mathcal{A}} Q_h(s_h, a)$
 observe state and reward $(s_{h+1}, r_h(s_h, a_h))$
 update dataset $\mathcal{D} \triangleq \mathcal{D} \cup \{(h, s_h, a_h, s_{h+1}, r_h(s_h, a_h))\}$
 end
end

where the exploration bonuses are based on the visitation counts and confidence parameter:

$$b(N) \triangleq cH \sqrt{\frac{L}{N}} \quad \text{where } L \triangleq \log\left(\frac{SAT}{\delta}\right) \quad (1)$$

This algorithm balances exploration and exploitation in two ways. The Q values are initialized to H , since this is their maximum possible value (number timesteps (H) $\times$ max reward per timestep (1.0)). This *optimistic* initialization promotes early exploration. In addition, the count-based bonuses help balance exploration and exploitation by providing a novelty bonus that incentivizes the algorithm to visit all states adequately and preventing the policy from being strictly greedy on previous experience. Together, these produce an algorithm that achieves good regret with very good sample complexity.

3 Regret Bound

We will define the Regret of UCB-VI in K episodes, with initial state fixed at s_1 :

$$R_T \triangleq K \cdot V_1^*(s_1) - \sum_{k=1}^K V_1^{\pi^k}(s_1)$$

Where V_1^* stands for the value of optimal policy and $V_1^{\pi^k}$ is the value of the policy sequence π^k

Theorem 1. *With probability at least $1 - \delta$, the regret of UCB-VI algorithm satisfies*

$$R_T \lesssim \sqrt{H^3 SATL} + H^3 S^2 AL^3$$

In the proof of Theorem 1, we will use the following notation:

Superscript k	the quantity used at episode k
$N_h^k(s, a, s')$	the number of (h, s, a, s') in $\mathcal{D}^k$ (dataset up to k -th episode)

$\hat{P}_h^k$	empirical transition probability
Q_h^k, V_h^k	computed value functions
$(s_h^k, a_h^k, s_{h+1}^k)$	trajectory played at k -th episode

To prove this theorem, we first need the following lemma.

Lemma 2. *With probability at least $1 - \delta$, we will have*

$$Q_h^k(s, a) \geq Q_h^*(s, a), \quad V_h^k(s) \geq V_h^* \quad \text{for all } (k, h, s, a)$$

i.e Q_h^k, V_h^k are upper confidence bound(UCB) of Q_h^*, V_h^* .

Proof We use induction to prove the lemma. For a fixed episode k , consider $h = H + 1, H, \dots, 1$. For $h = H + 1$, it holds that $V_{H+1}^k(s) \geq 0 = V_{H+1}^*(s)$ due to the algorithm setup. Assume $V_{h+1}^k(s) \geq V_{h+1}^*(s), \forall s \in \mathcal{S}$. If $(s, a) \notin \mathcal{D}^k$ the algorithm has not updated $Q_h^k(s, a)$, therefore, $Q_h^k(s, a) = H \geq Q_h^*(s, a)$. Otherwise,

$$Q_h^k(s, a) = \min\{H, r_h(s, a) + (\hat{P}_h^k V_{h+1}^k)(s, a) + b(N_h^k(s, a))\}$$

If the minimum is H , we have the induction. Otherwise,

$$\begin{aligned} Q_h^k(s, a) - Q_h^*(s, a) &= (\hat{P}_h^k V_{h+1}^k)(s, a) + b(N_h^k(s, a)) - (P_h V_{h+1}^*) \\ &= [\hat{P}_h^k (V_{h+1}^k - V_{h+1}^*)](s, a) + [(\hat{P}_h^k - P_h) V_{h+1}^*](s, a) + b(N_h^k(s, a)) \end{aligned}$$

By induction $[\hat{P}_h^k (V_{h+1}^k - V_{h+1}^*)](s, a) \geq 0$, and Hoeffding's inequality:

$$[(\hat{P}_h^k - P_h) V_{h+1}^*](s, a) + b(N_h^k(s, a)) \geq 0$$

Therefore, $Q_h^k(s, a) - Q_h^*(s, a) \geq 0$ and $V_h^k(s) = \max_{a \in \mathcal{A}} Q_h^k(s, a) \geq \max_{a \in \mathcal{A}} Q_h^*(s, a) = V_h^*(s)$ $\square$

Proof [Proof of Theorem 1] By Theorem 2, with high probability,

$$R_T = \sum_{k=1}^K (V_1^*(s_1) - V_1^{\pi^k}(s_1)) \leq \sum_{k=1}^K (V_1^k(s_1) - V_1^{\pi^k}(s_1))$$

$V_1^k(s_1)$ is the value maintained in the algorithm and $V_1^{\pi^k}(s_1)$ is the value function of π^k , which is the greedy policy with respect to Q^k . In general, we want to upper bound $V_h^k(s_h^k) - V_h^{\pi^k}(s_h^k)$ by a recursive form. Note that

$$V_h^k(s_h^k) - V_h^{\pi^k}(s_h^k) = (Q_h^k - Q_h^{\pi^k})(s_h^k, a_h^k)$$

because the action executed by π_k and the action used to update V_h^k are the same,

$$\pi^k(s_h^k) = \arg \max_{a \in \mathcal{A}} Q_h^k(s_h^k, a) = a_h^k$$

The difference between the Q functions can be decomposed as follows,

$$\begin{aligned} (Q_h^k - Q_h^{\pi^k})(s_h^k, a_h^k) &\leq (\hat{P}_h^k V_{h+1}^k - P_h V_{h+1}^{\pi^k})(s_h^k, a_h^k) + b(N_h^k(s_h^k, a_h^k)) \\ &= [\hat{P}_h^k - P_h] V_{h+1}^k(s_h^k, a_h^k) + [P_h (V_{h+1}^k - V_{h+1}^{\pi^k})](s_h^k, a_h^k) + b(N_h^k(s_h^k, a_h^k)) \\ &= [\hat{P}_h^k - P_h] V_{h+1}^*(s_h^k, a_h^k) + [\hat{P}_h^k - P_h] [V_{h+1}^k - V_{h+1}^*](s_h^k, a_h^k) \\ &\quad + [P_h (V_{h+1}^k - V_{h+1}^{\pi^k})](s_h^k, a_h^k) + b(N_h^k(s_h^k, a_h^k)) \end{aligned}$$

By Hoeffding's inequality, the first term above is bounded by:

$$\left[\hat{P}_h^k - P_h \right] V_{h+1}^*(s_h^k, a_h^k) \leq b(N_h^k(s_h^k, a_h^k))$$

Moreover, the second term can be related to the third, as follows:

$$\begin{aligned} & \left[\hat{P}_h^k - P_h \right] [V_{h+1}^k - V_{h+1}^*](s_h^k, a_h^k) \\ &= \sum_{s' \in \mathcal{S}} \left[\hat{P}_h^k(s' | s_h^k, a_h^k) - P_h(s' | s_h^k, a_h^k) \right] \cdot [V_{h+1}^k - V_{h+1}^*](s') \\ &\lesssim \sum_{s' \in \mathcal{S}} \left[\sqrt{\frac{P_h(s' | s_h^k, a_h^k) L}{N_h^k(s_h^k, a_h^k)}} + \frac{L}{N_h^k(s_h^k, a_h^k)} \right] \cdot [V_{h+1}^k - V_{h+1}^*](s') \\ &\lesssim \sum_{s' \in \mathcal{S}} \left[\frac{P_h(s' | s_h^k, a_h^k) L}{H} + \frac{HL}{N_h^k(s_h^k, a_h^k)} \right] \cdot [V_{h+1}^k - V_{h+1}^*](s') \\ &\lesssim \frac{1}{H} \left[P_h(V_{h+1}^k - V_{h+1}^*) \right](s_h^k, a_h^k) + \frac{H^2 SL}{N_h^k(s_h^k, a_h^k)} \\ &\lesssim \frac{1}{H} \left[P_h(V_{h+1}^k - V_{h+1}^{\pi_k}) \right](s_h^k, a_h^k) + \frac{H^2 SL}{N_h^k(s_h^k, a_h^k)}, \end{aligned}$$

where the first inequality holds with high probability, by Bernstein's inequality, the second inequality above used the AM-GM inequality, as demonstrated below:

$$\sqrt{\frac{P_h(s' | s_h^k, a_h^k) L}{N_h^k(s_h^k, a_h^k)}} = \sqrt{\frac{P_h(s' | s_h^k, a_h^k) L}{cH} \frac{cH}{N_h^k(s_h^k, a_h^k)}} \leq \frac{1}{2} \left(\frac{P_h(s' | s_h^k, a_h^k) L}{cH} + \frac{cH}{N_h^k(s_h^k, a_h^k)} \right)$$

and the third inequality removes the sum over $s' \in S$ in the first term, by making use of the fact that the “ P_h ” in $\left[P_h(V_{h+1}^k - V_{h+1}^*) \right](s_h^k, a_h^k)$ describes an “expectation” procedure, and thus implicitly involves a summation over s' . We also remove the sum over $s' \in S$ in the second term by trivially upper-bounding using $S := |\mathcal{S}|$ and $|[V_{h+1}^k - V_{h+1}^*](s')| \lesssim H$. The fourth and final inequality follows by observing that $V_{h+1}^{\pi_k} \leq V_{h+1}^*$ for each h, k .

(Proof modified above and continued below by Arnaud Fickinger and Chih-Yuan Chiu, scribes for Lecture 23, 4/13/2021.)

Let us briefly summarize our progress up to this point. We have established that:

$$\begin{aligned} (V_h^k - V_h^{\pi_k})(s_h^k) &= (Q_h^k - Q_h^{\pi_k})(s_h^k, a_h^k) \\ &\leq (\hat{P}_h^k - P_h) V_{h+1}^*(s_h^k, a_h^k) + (\hat{P}_h^k - P_h)(V_{h+1}^k - V_{h+1}^*)(s_h^k, a_h^k) \\ &\quad + \left[P_h(V_{h+1}^k - V_{h+1}^{\pi_k}) \right](s_h^k, a_h^k) + b(N_h^k(s_h^k, a_h^k)), \end{aligned}$$

and that:

$$\begin{aligned} (\hat{P}_h^k - P_h) V_{h+1}^*(s_h^k, a_h^k) &\leq b(N_h^k(s_h^k, a_h^k)), \\ \left[\hat{P}_h^k - P_h \right] [V_{h+1}^k - V_{h+1}^*](s_h^k, a_h^k) &\lesssim \frac{1}{H} \left[P_h(V_{h+1}^k - V_{h+1}^*) \right](s_h^k, a_h^k) + \frac{H^2 SL}{N_h^k(s_h^k, a_h^k)} \end{aligned}$$

Combining these inequalities, we thus have:

$$\begin{aligned} & (V_h^k - V_h^{\pi_k})(s_h^k) \\ &\leq (\hat{P}_h^k - P_h) V_{h+1}^*(s_h^k, a_h^k) + (\hat{P}_h^k - P_h)(V_{h+1}^k - V_{h+1}^*)(s_h^k, a_h^k) \end{aligned}$$

$$\begin{aligned}
& + \left[P_h(V_{h+1}^k - V_{h+1}^{\pi_k}) \right] (s_h^k, a_h^k) + b(N_h^k(s_h^k, a_h^k)) \\
& \leq \left(1 + \frac{1}{H} \right) \cdot \left[P_h(V_{h+1}^k - V_{h+1}^{\pi_k}) \right] (s_h^k, a_h^k) + \frac{H^2 SL}{N_h^k(s_h^k, a_h^k)} + 2b(N_h^k(s_h^k, a_h^k)).
\end{aligned}$$

We wish to establish a recursive relationship between terms of the form $(V_h^k - V_h^{\pi_k})$'s over the index $h \in [H]$, so that we can upper bound $(V_1^k - V_1^{\pi_k})(s_1)$. To this end, we introduce the difference term " $(V_{h+1}^k - V_{h+1}^{\pi_k})(s_{h+1}^k)$ " into the above chain of inequalities by defining:

$$\begin{aligned}
\xi_h^k &:= \left[P_h(V_{h+1}^k - V_{h+1}^{\pi_k}) \right] (s_h^k, a_h^k) - (V_{h+1}^k - V_{h+1}^{\pi_k})(s_{h+1}^k), \\
\zeta_h^k &:= \frac{H^2 SL}{N_h^k(s_h^k, a_h^k)}, \\
b_h^k &:= b(N_h^k(s_h^k, a_h^k)).
\end{aligned}$$

The above bound for $(V_h^k - V_h^{\pi_k})(s_h^k)$ then becomes:

$$\begin{aligned}
& (V_h^k - V_h^{\pi_k})(s_h^k) \\
& \leq \left(1 + \frac{1}{H} \right) \cdot \left[P_h(V_{h+1}^k - V_{h+1}^{\pi_k}) \right] (s_h^k, a_h^k) + \frac{H^2 SL}{N_h^k(s_h^k, a_h^k)} + 2b(N_h^k(s_h^k, a_h^k)) \\
& \leq \left(1 + \frac{1}{H} \right) \cdot \left[(V_{h+1}^k - V_{h+1}^{\pi_k})(s_{h+1}^k) + \xi_h^k \right] + \zeta_h^k + 2b_h^k.
\end{aligned}$$

Unrolling this recursion, we arrive at the following bound for $(V_1^k - V_1^{\pi_k})(s_1)$:

$$\begin{aligned}
(V_1^k - V_1^{\pi_k})(s_1) & \leq \left(1 + \frac{1}{H} \right) \cdot (\xi_1^k + \zeta_1^k + 2b_1^k) + \left(1 + \frac{1}{H} \right)^2 \cdot (\xi_2^k + \zeta_2^k + 2b_2^k) \\
& \quad + \cdots + \left(1 + \frac{1}{H} \right)^H \cdot (\xi_H^k + \zeta_H^k + 2b_H^k)
\end{aligned}$$

Since $\left(1 + \frac{1}{h} \right)^h$, for each $h \in [H]$, increases in h and uniformly bounded above, e.g., by the constant 3. Thus, to upper bound the regret $R_T \leq \sum_{k=1}^K [V_1^k(s_1) - V_1^{\pi_k}(s_1)]$ (the objective as given in the theorem statement), it suffices to bound the sums of $\xi_h^k, \zeta_h^k, b_h^k$ over $h \in [H]$ and $k \in [K]$. We do so below.

- Bounding ξ_h^k :

Bounding ξ_h^k requires use of the Azuma-Hoeffding inequality. More precisely, since $V_h^k \in \left(0, \frac{R_{\max}}{1-\gamma} \right)$ for each $h \in [H]$ and $k \in [K]$, we have by definition of ξ_h^k :

$$P \left(\left| \sum_{k=1}^K \sum_{h=1}^H \xi_h^k \right| \geq t \right) \leq \exp \left(- \frac{2t^2}{HK \left(\frac{R_{\max}}{1-\gamma} \right)^2} \right), \quad \forall t \geq 0.$$

Substituting:

$$t = \sqrt{\frac{HK}{2} \cdot \left(\frac{R_{\max}}{1-\gamma} \right)^2} \cdot \sqrt{\log \frac{1}{\delta}} = O(\sqrt{HK}),$$

we find that with high probability:

$$\sum_{k=1}^K \sum_{h=1}^H \xi_h^k \lesssim \sqrt{HK}$$

- Bounding ζ_h^k :

Note that:

$$\begin{aligned}
\sum_{k=1}^K \sum_{h=1}^H \zeta_h^k &= H^2 SL \cdot \sum_{k=1}^K \sum_{h=1}^H \frac{1}{N_h^k(s_h^k, a_h^k)} \\
&\lesssim H^2 SL \cdot \sum_{(h,s,a) \in [H] \times \mathcal{S} \times \mathcal{A}} \sum_{i=1}^{N_h^k(s,a)} \frac{1}{i} \\
&\lesssim H^2 SL \cdot \sum_{(h,s,a) \in [H] \times \mathcal{S} \times \mathcal{A}} \log(N_h^k(s, a)) \\
&\lesssim H^2 SL \cdot HSA \log T,
\end{aligned}$$

where the first inequality follows by rearranging the double sum so that the quantity $\frac{1}{N_h^k(s_h^k, a_h^k)}$ is summed over each state-action pair $(s, a) \in \mathcal{S} \times \mathcal{A}$ (i.e., the pigeonhole principle), the second inequality follows by bounding the sum of reciprocals of the integers, and the third inequality follows by trivially upper-bounding $N_h^k(s, a)$ by T .

- Bounding b_h^k :

Recall that $b_h^k := b(N_h^k(s_h^k, a_h^k)) = cH \sqrt{\frac{L}{N_h^k(s_h^k, a_h^k)}}$. Applying the pigeonhole principle again, we have:

$$\begin{aligned}
\sum_{k=1}^K \sum_{h=1}^H b_h^k &\lesssim H\sqrt{L} \cdot \sum_{k=1}^K \sum_{h=1}^H \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k)}} \\
&= H\sqrt{L} \cdot \sum_{(h,s,a) \in [H] \times \mathcal{S} \times \mathcal{A}} \sum_{i=1}^{N_h^k(s,a)} \frac{1}{\sqrt{i}} \\
&= H\sqrt{L} \cdot \sum_{(h,s,a) \in [H] \times \mathcal{S} \times \mathcal{A}} \sum_{i=1}^{N_h^k(s,a)} \frac{1}{\sqrt{i}} \\
&\lesssim H\sqrt{L} \cdot \sum_{(h,s,a) \in [H] \times \mathcal{S} \times \mathcal{A}} \sqrt{N_h^k(s, a)} \\
&\lesssim H\sqrt{L} \cdot HSA\sqrt{T}.
\end{aligned}$$

Summing up the bounds for ξ_h^k , ζ_h^k and b_h^k gives the desired claim. □

References

- [1] M. G. Azar, I. Osband, and R. Munos, “Minimax regret bounds for reinforcement learning,” in *International Conference on Machine Learning*. PMLR, 2017, pp. 263–272.